Name _____ Date _____

Applying Knowledge
Section 2.3

Use with textbook pages 64–67.

The number game with atoms and ions

1. Complete the following sentences using the terms in parentheses.
   - **(a)** The atomic _____ (number/mass) of an element is the same as the number of protons in the nucleus of an atom.
   - **(b)** An _____ (atom/ion) of an element has the same number of protons as electrons.
   - **(c)** A positively charged ion has _____ (lost/gained) electrons.
   - **(d)** A negatively charged ion has _____ (lost/gained) electrons.

2. Complete the following table. Some answers are provided to help guide you. You can refer to the Bohr model chart on page 32 and the periodic table on page 202.

Element name	Atomic number	Ion charge	Atom or ion?	Number of protons	Number of electrons
beryllium	4	2+	ion	4	2
	11	0	atom		
		0		18	18
chlorine		0			
	7	3–			10
calcium		0			
		2–		16	
	3	+			
		3+		13	

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